

Su-un Lee

PhD Candidate at The University of Chicago

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Education

Ph.D. in Quantum Science and Engineering

Sep 2022 – Present

University of Chicago, Chicago, IL

Advisor: [Prof. Liang Jiang](#)

B.S. in Physics & B.S. in Mathematical Sciences

Mar 2016 – Aug 2022*

Seoul National University, Seoul, South Korea

Summa Cum Laude

* Military service [Mar. 2019 – Oct. 2020].

Work Experience

Returning Intern: Research Scientist

Jun 2026 – Sep 2026

IBM Research, Chicago, IL

Mentor: Kunal Sharma

Quantum Computing Theory Research Scientist Intern

Jun 2025 – Sep 2025

IBM Thomas J. Watson Research Center, Yorktown Heights, NY

Mentors: Kunal Sharma, Abhinav Deshpande, and Kevin Smith

Featured in [Engineering the summer: Advancing quantum computing at IBM](#)

Research Highlights

Techniques for Early Fault-tolerant Quantum Devices

- *Quantum error mitigation assisting logical quantum information processing*: Developed a method to assist logical quantum information processing with probabilistic error cancellation.
- *Benchmarking logical quantum information processing*: Developed efficient benchmarking methods for logical states and logical channels.
- *Benchmarking dynamical quantum circuits*: Developed methods for benchmarking quantum circuits with mid-circuit measurements and reset channels.

Opportunities and Limitations of Near-term Quantum Computers

- *Classical tractability of quantum advantage experiments*: Benchmarked state-of-the-art classical simulation methods, including sparse Pauli propagation, 2D tensor network, and belief propagation, assisting two quantum advantage experiments at IBM.
- *Classical simulation of noisy random quantum circuits*: Developed classical simulation methods for random circuits with generic noise.

- *Unitary-assisted quantum cooling*: Developed a method to enhance quantum cooling with coherent control.
- *Algorithmic cooling with qutrits*: Proposed a method to enhance algorithmic cooling which leverages the three-level system (demonstration in progress).

Learning classically representable quantum states and processes

- *Efficient learning of sparse quantum states*: Developed efficient algorithms for learning entangled quantum states from single-qubit measurements.

Information Processing and Complexity in Random Quantum Circuits

- *Information processing in noisy random circuits*: Discovered a universal information spreading behavior that is intrinsic to noisy random circuits.

Technical Skills & Experience

AI for Scientific Discovery

- Organize and host a recurring group meeting focused on developing AI-assisted workflows for research.
- Harness engineering for integrating AI into research pipelines.
- Verification of AI-generated mathematical proofs with LEAN.

Classical Simulation of Quantum Systems / Quantum Circuits

- Developed a pipeline for Projected Entangled Pair States (PEPS) and Belief-Propagation (BP) simulations (being actively used in multiple projects in IBM Research).
- Large-scale Matrix Product State (MPS) and Matrix Product Operator (MPO) simulation for pure and mixed states.
- Large-scale stabilizer simulation for pure and mixed states.

Machine Learning and Deep Learning

- Construct and train deep neural networks with PyTorch and TensorFlow, accelerated with Multi-GPU programming.
- Developed automatic hyperparameter optimization routines with Optuna.

High-Performance Computing (HPC)

- Cloud computing and multi-GPU programming supported by NVIDIA Academic Grant Program.
- Midway, Research Computing Center, University of Chicago
- [Nurion](#) & [Neuron](#), Korea Institute of Science and Technology Information

Programming

- Programming languages: Python, Julia, etc.
- Version control: Git and GitHub

Awards & Honors

- **Kwanjeong Educational Foundation Fellowship** 2022 – 2027
- **Dean’s List**, College of Natural Sciences, Seoul National University 2018, 2020
- **Excellence Award**, Outstanding Report Contest, Seoul National University 2018
- **SNU Development Fund Scholarship**, Seoul National University 2017
- **Eminence Scholarship**, Seoul National University 2016, 2018, 2019
- **Merit-based Scholarship**, Seoul National University 2017, 2021

Publications and Preprints

In reverse chronological order. See also [Google Scholar](#)

- [1] **Su-un Lee**, Liang Jiang, and Kunal Sharma, Learning sparse quantum states from single-qubit measurements. (2026), arXiv:2609.35648 [quant-ph].
- [2] Samantha V Barron, Bradley Mitchell, Vinay Tripathi, Francesco Grieco, Ilan Rosen, Francesca Pietracaprina, Davide Materia, Alireza Seif, Darvin Wanisch, Ramón L Panadés-Barrueta, Ewout van den Berg, Jay-U Chung, Andrew Eddins, Sam Ferracin, Guillermo García-Pérez, John Goold, Luke CG Govia, Holger Haas, Ian Hincks, Jesse C Hoke, Zoë Holmes, **Su-un Lee**, Youngseok Kim, Swarnadeep Majumder, Sabrina Maniscalco, Simone Montangero, Daniel Puzzuoli, Tomaž Prosen, James Raftery, Ricardo Rivera Cardoso, Max Rossmannek, Manuel Rudolph, Brendan Saxberg, Liran Shirizly, Karthik Siva, Joshua Skanes-Norman, Iliaria Siloi, Kevin C Smith, Boris Sokolov, Maika Takita, Yanting Teng, Mao Tian Tan, Joseph Tindall, Zoltán Zimborás, Matteo AC Rossi, Minh C Tran, Sergei N Filippov, Abhinav Kandala, Observable Estimation in the Absence of Classical Verification. (2026), arXiv:2607.25998 [quant-ph].
- [3] Chia-Tung Chu, **Su-un Lee**, Han Zheng, Senrui Chen, Bibek Pokharel, Alireza Seif, and Liang Jiang, Learning Mid-circuit Measurement Backaction from Three Repeated Measurements. (2026), arXiv:2606.00433 [quant-ph].
- [4] William Kirby, Bibek Pokharel, Javier Robledo Moreno, Kevin C. Smith, Sergey Bravyi, Abhinav Deshpande, Constantinos Evangelinos, Bryce Fuller, James R. Garrison, Ben Jaderberg, Caleb Johnson, Petar Jurcevic, **Su-un Lee**, Simon Martiel, Mario Motta, Seetharami Seelam, Oles Shtanko, Kevin J. Sung, Minh Tran, Vinay Tripathi, Kazuhiro Seki, Kazuya Shinjo, Han Xu, Lukas Broers, Tomonori Shirakawa, Seiji Yunoki, Kunal Sharma, and Antonio Mezzacapo, Observation of Improved Accuracy over Classical Sparse Ground-State Solvers using a Quantum Computer. (2026), arXiv:2603.03496 [quant-ph].
- [5] **Su-un Lee***, Ming Yuan*, Senrui Chen, Kento Tsubouchi, and Liang Jiang, Efficient benchmarking of logical magic states *Phys. Rev. Lett.* **136**, 050602 (2026).
- [6] Han Zheng, Chia-Tung Chu, Senrui Chen, Argyris Giannisis Manes, **Su-un Lee**, Sisi Zhou, and Liang Jiang, Efficient learning of logical noise from syndrome data. (2026), arXiv:2601.22286 [quant-ph].

- [7] **Su-un Lee**, Soumik Ghosh, Changhun Oh, Kyungjoo Noh, Bill Fefferman, and Liang Jiang, Classical simulation of noisy random circuits from exponential decay of correlation. (2025), arXiv:2510.06328 [quant-ph]. (Submitted to *Phys. Rev. Lett.*).
- [8] Yifan F. Zhang*, **Su-un Lee***, Liang Jiang, and Sarang Gopalakrishnan, Classically sampling noisy quantum circuits in quasi-polynomial time under approximate Markovianity (2025), arXiv:2510.06324 [quant-ph].
- [9] **Su-un Lee**, Changhun Oh, Yat Wong, Senrui Chen, and Liang Jiang, Universal spreading of conditional mutual information in noisy random circuits. *Phys. Rev. Lett.* **133**, 200402 (2024).

* Equal contribution

Presentations

Invited Talks

- IBM Quantum Computing PIC Seminar, IBM Research, IL Sep, 2026
- IBM Quantum Computing PIC Seminar, IBM T. J. Watson Research Center, NY Mar, 2026
- Google Quantum AI Tensor Network Seminar, Google Quantum AI, CA Jan, 2026
- IBM Quantum Computing PIC Seminar, IBM T. J. Watson Research Center, NY Oct, 2025
- IBM Quantum Computing PIC Seminar, IBM T. J. Watson Research Center, NY Sep, 2025
- Quantum Information Science and Engineering Seminar, University of Chicago, IL Dec, 2024

Contributed Talks

- Global Physics Summit 2026, Denver, CO March, 2026
- Korea Quantum Information Science Conference (K-QIS), Seoul, South Korea Nov, 2025
- Global Physics Summit 2025, Anaheim, CA March, 2025
- APS March Meeting 2024, Minneapolis, MN March, 2024

Academic Service

Program Committee

- 2nd Workshop on HPC/AI Integration with Quantum Computing/Networking (HAIQ 2026)

Journal Referee services

- *Physical Review A* (7 times)
- *PRX Quantum* (2 times)
- *Physical Review Research* (1 time)

- *Physical Review Applied* (2 times)
- *APS Open Science* (1 time)
- *Quantum* (1 time)

Conference Reviewer

- *Theory of Quantum Computation, Communication and Cryptography (TQC 2026)*
- *29th Annual Conference on Quantum Information Processing (QIP 2026)*
- *25th Asian Quantum Information Science Conference (AQIS 2025)*
- *Theory of Quantum Computation, Communication and Cryptography (TQC 2025)*
- *28th Annual Conference on Quantum Information Processing (QIP 2025)*

Teaching Experience

University of Chicago

- Guest Lecturer – Advanced Quantum Information and Computation Winter 2025
- Teaching Assistant – Advanced Quantum Information and Computation Winter 2025
- Teaching Assistant – Intermediate Quantum Engineering Autumn 2023

Seoul National University

- Teaching Assistant – Foundation of Physics 1 Spring 2022
- Teaching Assistant – Foundation of Physics 1 Spring 2021
- Teaching Assistant – Basic Calculus 1 Spring 2018

References

Liang Jiang

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University of Chicago
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Bill Fefferman

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Kunal Sharma

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